

TASK 11 - NumPy Vectorization and Broadcasting

1. Task Overview

Perform arithmetic operations on complete NumPy arrays and explore broadcasting between arrays of compatible shapes.

2. Objective

Understand efficient numerical computation without explicit Python loops.

3. Tools

Python, NumPy, and Jupyter Notebook.

4. Vectorization - Basic Concept

Vectorization means applying an operation to an entire NumPy array at once instead of processing each element with an explicit Python loop.

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])
arr + 5
arr * 2
arr ** 2
```

5. Vectorized Arithmetic

NumPy applies arithmetic element by element across complete arrays.

```
a = np.array([10, 20, 30, 40])
b = np.array([1, 2, 3, 4])

a + b
a - b
a * b
a / b
```

6. Scalar Broadcasting

A scalar can be applied to every element of an array automatically.

```
marks = np.array([60, 70, 80, 90])
marks + 5
# [65 75 85 95]
```

7. Broadcasting - Meaning

Broadcasting is NumPy's mechanism for performing operations between arrays with compatible shapes without manually repeating values.

8. Broadcasting Example 1 - Scalar

A scalar broadcasts to every array element.

```
arr = np.array([10, 20, 30])
arr + 5
# [15 25 35]
```

9. Broadcasting Example 2 - Row Vector

A 1D row array can be broadcast across each row of a compatible 2D matrix.

```
matrix = np.array([[10,20,30],
                  [40,50,60],
                  [70,80,90]])
row = np.array([1,2,3])
matrix + row
```

10. Broadcasting Example 3 - Column Vector

A column-shaped array can be broadcast down the rows.

```
column = np.array([[10],[20],[30]])
matrix + column
```

11. Broadcasting Rules

Dimensions are compared from the right. They are compatible when they are equal or when one dimension is 1. Missing dimensions are treated as 1.

Situation	Compatible?
Same dimensions	Yes
One dimension is 1	Yes
Scalar with array	Yes
Different non-1 dimensions	No

12. Student Marks Practical

The same adjustment can be applied to all student marks without writing a loop.

```
student_marks = np.array([
    [78,82,91],
    [65,74,80],
    [88,90,95],
    [55,68,72]
])
updated_marks = student_marks + 5
print(updated_marks)
```

13. Iris-Style Numerical Data

Vectorized operations can also be applied to numerical feature data.

```
iris = np.array([
    [5.1,3.5,1.4,0.2],
    [4.9,3.0,1.4,0.2],
    [6.2,3.4,5.4,2.3]
])
scaled = iris * 10
```

14. Loop-Based Calculation

A traditional approach processes values one by one using an explicit Python loop.

```
values = np.array([10,20,30,40,50])
result = []
for x in values:
    result.append(x * 2)
```

15. Vectorized Calculation

The same calculation can be performed directly on the complete array.

```
values = np.array([10,20,30,40,50])
result = values * 2
```

16. Loop-Based vs Vectorized Calculation

Both approaches can produce the same result. Vectorized NumPy syntax is concise and designed for array-based numerical computation.

Loop-Based	Vectorized
Explicit Python loop	Complete-array operation
More code	Less code
Element-by-element	Array-level operation

17. Why Vectorization is Useful

It reduces explicit loop code and provides a convenient approach for numerical array operations in data analysis.

18. Common Mistakes

1. Assuming every shape is compatible.
2. Forgetting that dimensions are compared from the right.
3. Using incompatible row and column shapes.
4. Adding unnecessary loops when direct NumPy arithmetic is available.
5. Confusing vectorization with broadcasting.

19. Interview Questions and Answers

What is vectorization? Performing an operation on an entire NumPy array instead of explicitly looping through elements.

What is broadcasting? A NumPy mechanism that allows operations between compatible arrays of different shapes.

Why is vectorization useful? It makes numerical code concise and supports efficient array computation.

Can a scalar be broadcast? Yes. A scalar can be applied to every element of an array.

Give one broadcasting example. Adding a 1D row array to each row of a compatible 2D matrix.

What makes two dimensions compatible? They are equal, or one dimension is 1; missing dimensions are treated as 1.

Difference between vectorization and broadcasting? Vectorization describes array-level computation; broadcasting describes shape alignment for an operation.

20. Quick Revision

```
Vectorization -> operation on a complete array
Broadcasting  -> compatible shapes work together
Scalar        -> can broadcast to an array
Row vector    -> can broadcast across rows
Column vector -> can broadcast down rows
Loop          -> explicit element-by-element processing
```

21. Complete Mini Program

```
import numpy as np

arr = np.array([10,20,30,40,50])
print("Add 5:", arr + 5)
print("Multiply by 2:", arr * 2)

matrix = np.array([[10,20,30],
                   [40,50,60],
                   [70,80,90]])

row = np.array([1,2,3])
column = np.array([[10],[20],[30]])

print(matrix + row)
print(matrix + column)

marks = np.array([[78,82,91],
                  [65,74,80],
                  [88,90,95],
                  [55,68,72]])

print(marks + 5)
```

22. Expected Learning Outcome

After completing this task, the learner should be able to perform vectorized arithmetic, understand scalar and array broadcasting, recognize compatible shapes, compare loop-based and vectorized calculations, and apply these concepts to numerical datasets.

23. Top 5 Skills

1. NumPy
2. Vectorization
3. Broadcasting
4. Numerical Data Processing

5. Python Data Analysis

24. Feedback

The task was very helpful for understanding how NumPy vectorization and broadcasting make numerical calculations more efficient and concise. I learned how to perform arithmetic operations on complete arrays, apply scalar and array broadcasting, and work with compatible shapes. The practical student marks examples and loop-versus-vectorized comparison improved my understanding of efficient numerical data processing.

25. Submission Checklist

- ✓ Vectorized arithmetic examples
- ✓ At least three broadcasting examples
- ✓ Scalar broadcasting
- ✓ Row broadcasting
- ✓ Column broadcasting
- ✓ Student marks practical
- ✓ Iris-style numerical example
- ✓ Loop-based vs vectorized comparison
- ✓ Interview questions
- ✓ Python script and Jupyter Notebook
- ✓ README documentation

26. Final Summary

Vectorization allows NumPy to perform operations on complete arrays without explicit Python loops. Broadcasting allows compatible arrays of different shapes to participate in the same operation. Together, these concepts are important for efficient numerical computing and data analysis.

TASK 11 - FULL AND COMPLETE: COMPLETED ✓